Literature Review

❖ Introduction

Disaster management communication plays a crucial role in minimizing loss of life and property during natural calamities such as floods, landslides, earthquakes, and cyclones. Early Warning Systems (EWS) are designed to disseminate timely alerts to the public to enable preventive and protective action. According to Mileti and Sorensen (1990), the effectiveness of warning systems depends not only on speed of dissemination but also on clarity, consistency, and credibility of the message. Similarly, Lindell and Perry (2012) proposed the Protective Action Decision Model (PADM), which explains that individuals take protective action only when they understand the risk and trust the source of communication.

In linguistically diverse regions like Kerala, India, disaster communication faces additional challenges due to language barriers, literacy levels, and accessibility issues. Although government agencies such as NDMA and IMD provide SMS and broadcast-based alerts, these systems often lack multilingual adaptability, accessibility support, and mechanisms to measure public understanding. Therefore, research indicates the need for inclusive, structured, and feedback-driven disaster communication systems.

❖ Existing System Limitations

2.1 Language Barriers

Eisenman et al. (2007) found that non-native speakers and migrant populations are significantly less likely to understand emergency warnings issued in a dominant language. In multilingual regions, issuing alerts in only one language reduces comprehension and delays response time.

Kerala, which hosts a large migrant workforce, faces similar risks when disaster warnings are issued primarily in English or Malayalam without multilingual flexibility.

2.2 Lack of Accessibility

Research in Universal Access in the Information Society (Lazar et al., 2017) highlights that elderly individuals and visually impaired users struggle with text-only alerts. Most disaster communication systems lack integrated Text-to-Speech (TTS) support, making them inaccessible to vulnerable populations.

2.3 Inconsistent Messaging and Misinformation

Sutton et al. (2014) analyzed crisis communication on Twitter and found that inconsistent and repeated messaging across different platforms creates confusion. Social media platforms often spread rumors, which can escalate panic and misdirect resources.

2.4 Absence of Comprehension Measurement

Studies on FEMA's Wireless Emergency Alerts (Bean et al., 2015) reveal that most alert systems measure reach but not understanding. There is no structured mechanism to assess whether recipients clearly understood the warning message.

2.5 Lack of Structured Geo-Filtering and Metadata

Many systems broadcast alerts broadly without allowing users to filter by district, severity level, or disaster type. This can result in information overload and reduced attention to critical alerts.

❖ Comparative Analysis of Existing Systems

Feature	NDMA SMS	FEMA WEA	Social Media	Proposed System
Geo-targeted alerts	Partial	Yes	No	Yes
Multilingual AI translation	Limited	Limited	No	Yes
Text-to-Speech support	No	No	No	Yes
Severity-level filtering	Basic	Basic	No	Yes
Feedback tracking	No	No	No	Yes
Structured metadata	Limited	Limited	No	Yes
Misinformation control	**Moderate**	**Strong**	**Weak**	**Strong**

Figure 1

From the comparison, it is evident that existing systems focus primarily on message broadcasting rather than accessibility, multilingual adaptability, and comprehension analytics.

Figure-1 presents a comparative analysis of major existing disaster communication systems, including NDMA SMS alerts (India), FEMA Wireless Emergency Alerts (USA), and social media platforms, alongside the proposed system.

The comparison highlights key functional parameters such as geo-targeting, multilingual support, accessibility features, metadata structuring, misinformation control, and feedback tracking. While FEMA's WEA provides strong geo-targeting and institutional control, it lacks multilingual AI translation, accessibility features like Text-to-Speech, and public comprehension measurement.

Similarly, NDMA SMS alerts provide basic dissemination but offer limited filtering and no structured feedback mechanisms.

Social media platforms enable rapid information spread; however, they lack official verification controls, structured metadata, and misinformation management, which can lead to confusion during emergencies.

In contrast, the proposed system integrates geo-targeted filtering, AI-based multilingual translation, Text-to-Speech accessibility, structured metadata tagging, and a feedback-based awareness tracking mechanism.

This comparison demonstrates that existing systems primarily emphasize message broadcasting, whereas the proposed platform extends beyond dissemination to focus on accessibility, clarity, structured communication, and measurable public understanding.

❖ Identified Research Gap

The literature indicates that:

- Early warning systems prioritize speed over inclusivity.
- Multilingual disaster communication remains limited.
- Accessibility features such as Text-to-Speech are rarely integrated.
- No mainstream system includes measurable comprehension feedback.

There is a clear need for a structured, multilingual, location-based disaster alert platform that integrates AI-powered translation, accessibility features, and public awareness tracking within a unified framework.

❖ Conclusion

The reviewed literature highlights that effective disaster communication requires clarity, inclusivity, accessibility, and measurable public engagement. Existing systems such as NDMA SMS alerts, FEMA WEA, and social media-based updates provide rapid dissemination but lack structured multilingual support, accessibility-driven audio features, and comprehension feedback mechanisms.

Therefore, developing a location-based, multilingual disaster alert platform with AI-powered translation, Text-to-Speech functionality, and awareness tracking directly addresses the identified research gaps. Such a system aligns with SDG 11 (Sustainable Cities and Communities), SDG 13 (Climate Action), and SDG 16 (Strong Institutions), contributing to resilient and inclusive disaster response infrastructure.

❖ References

1. *Mileti, D. S., & Sorensen, J. H. (1990). Communication of Emergency Public Warnings: A Social Science Perspective and State-of-the-Art Assessment. Oak Ridge National Laboratory. Official Report (PDF): https://apps.dtic.mil/sti/pdfs/ADA268812.pdf*

2. *Lindell, M. K., & Perry, R. W. (2012). The Protective Action Decision Model: Theoretical Modifications and Additional Evidence. Risk Analysis, 32(4), 616–632. Publisher Link: https://onlinelibrary.wiley.com/doi/10.1111/j.1539-6924.2011.01647.x*

3. *Eisenman, D. P., Cordasco, K. M., Asch, S., Golden, J. F., & Glik, D. (2007). Disaster Planning and Risk Communication With Vulnerable Communities. American Journal of Public Health, 97(S1). Link: https://ajph.aphapublications.org/doi/10.2105/AJPH.2005.084335*

4. *Sutton, J., Spiro, E. S., Johnson, B., Fitzhugh, S., Gibson, B., & Butts, C. T. (2014). Warning Tweets: Serial Transmission of Messages During the Warning Phase of a Disaster Event. Information, Communication & Society, 17(6), 765–787. Link: https://www.tandfonline.com/doi/abs/10.1080/1369118X.2013.862561*

5. *Bean, H., Liu, B. F., Madden, S., Sutton, J., Wood, M. M., & Mileti, D. S. (2015). Comprehensive Testing of Wireless Emergency Alerts. Natural Hazards Review, 16(2). Link: https://ascelibrary.org/doi/10.1061/(ASCE)NH.1527-6996.0000146*

6. *Lazar, J., et al. (2017). Ensuring Digital Accessibility Through Process and Policy. Morgan Kaufmann Link: https://www.sciencedirect.com/book/9780128007107/ensuring-digital-accessibility-through-process-and-policy*

7. *United Nations Office for Disaster Risk Reduction (UNDRR) (2022). Early Warning Systems for All: Executive Action Plan 2023–2027. Official UN Publication: https://www.undrr.org/publication/early-warning-systems-all*